CHAPTER 3

Vector calculation and scalar product

Chapter path

Objectives. Choose between projection, coordinates and angle; calculate a length; characterise orthogonality and a locus.
Prerequisites. Vectors, Chasles' relation, Pythagoras, cosine in a right-angled triangle, orthonormal coordinates.

Yearly reference point. 6-15 October: 8 hours

Diagnostic check: six questions, 10 minutes

Answer without looking at your notes. Indicate the questions for which help is needed. This diagnostic test is not graded.

1. For $A(1;2)$ and $B(4;-2)$, calculate $\overrightarrow{AB}$.

2. Calculate the norm of $(3;-4)$.

3. Simplify $\overrightarrow{AB}+\overrightarrow{BC}$.

4. Expand $(a+b)^2$.

5. In a right triangle, an angle has adjacent side 3 and hypotenuse 5. Find its cosine.

6. The midpoint of $A(-2;1)$ and $B(4;3)$?

After the diagnostic check. Go over the flagged concepts with the teacher; redo a similar question, then explain the rule used.

Oral work and logic. Compare, on the same problem, coordinates and coordinate-free vector calculation. Justify why your method uses the data most directly.

Investigate before formalising

Construct three points M such that $\overrightarrow{MA} \cdot \overrightarrow{MB} = 0$ for a segment AB of 6 cm. Conjecture the locus then produce a proof that includes A and B.

Organisation. Individual work 5 min, discussion in pairs 5 min, whole-class sharing 5 min. Keep a record of an attempt, even an unfinished one. A conjecture must then be justified.

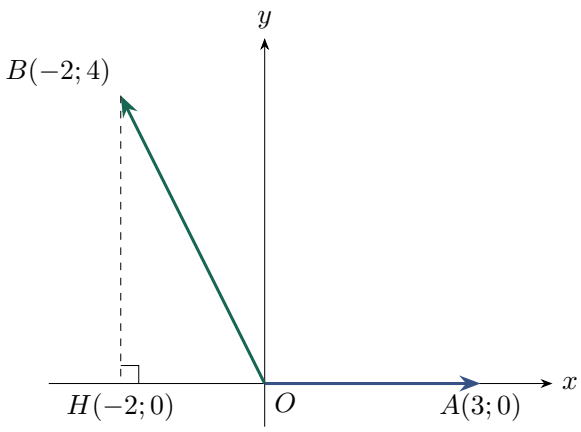
3.1 Projection and angle

Definition and results

For two non-zero vectors $\vec{u}, \vec{v}$, their scalar product is the real number $\vec{u} \cdot \vec{v} = \|\vec{u}\| \, \|\vec{v}\| \cos\theta$, where $\theta \in [0; \pi]$ is their non-oriented angle. With a zero vector, the product is 0. If H is the projection of B on (OA), $\overrightarrow{OA} \cdot \overrightarrow{OB}$ is $OA \times OH$ when H is on the ray $[OA)$, and its opposite if not.

Status and justification. Definition; the equivalence with projection follows from the cosine in right-angled triangles, with a negative algebraic projection for an obtuse angle.

Worked example. For two vectors with norms 3 and 4 at an angle of 60 °, the product is $\dfrac{3 \times 4 \times 1}{2} = 6$. With 120 °, it is -6. If $A(3; 0)$ and $B(-2; 4)$, the projection is $H(-2; 0)$ and the product is -6.



The projection H is on the negative side of the horizontal axis. Thus $\overrightarrow{OA} \cdot \overrightarrow{OB} = 3 \times (-2) = -6$.

Watch out

The result is a number, not a vector. A right angle corresponds to 0, not 1.

Immediate application. Two vectors have norms 2 and 5 and form 90 °, then 180 °. Calculate the products.

3.2 Coordinates, Norm and Orthogonality

Definition and results

In an orthonormal basis, for $\vec{u} = (x; y)$ and $\vec{v} = (x\prime; y\prime)$, $\vec{u} \cdot \vec{v} = xx\prime + yy\prime$ and $\|\vec{u}\|^2 = x^2 + y^2$. The vectors are orthogonal if and only if their scalar product is zero. For a basis $(\vec{i}, \vec{j})$ orthonormal, $x = \vec{u} \cdot \vec{i}$ and $y = \vec{u} \cdot \vec{j}$.

Status and justification. Properties proved using bilinearity: expand $(x\vec{i} + y\vec{j}) \cdot (x\prime\vec{i} + y\prime\vec{j})$ and use $\vec{i} \cdot \vec{i} = \vec{j} \cdot \vec{j} = 1$, $\vec{i} \cdot \vec{j} = 0$.

Worked example. $\vec{u} = (2; 3)$, $\vec{v} = (3; -2)$: product 6 − 6 = 0. The two vectors are orthogonal. Each has norm $\sqrt{13}$. The product of (2; 3) with (1; 0) is 2: we recover the x-coordinate.

Watch out

The usual formula assumes an orthonormal basis. Collinearity and orthogonality are different properties.

Immediate application. For which real t are the vectors $(t; 2)$ and $(3; -6)$ orthogonal?

3.3 Vector calculation and identities

Definition and results

The scalar product is symmetric and bilinear: $\vec{u} \cdot (\vec{v} + \vec{w}) = \vec{u} \cdot \vec{v} + \vec{u} \cdot \vec{w}$ and $(\lambda\vec{u}) \cdot \vec{v} = \lambda(\vec{u} \cdot \vec{v})$. Thus $\|\vec{u} \pm \vec{v}\|^2 = \|\vec{u}\|^2 \pm 2\vec{u} \cdot \vec{v} + \|\vec{v}\|^2$. We deduce $\vec{u} \cdot \vec{v} = \dfrac{\|\vec{u} + \vec{v}\|^2 - \|\vec{u}\|^2 - \|\vec{v}\|^2}{2}$.

Status and justification. Bilinearity assumed from the geometric definition; symmetry comes from the cosine. The identities are proved by expanding $(\vec{u} \pm \vec{v}) \cdot (\vec{u} \pm \vec{v})$.

Worked example. If $\|\vec{u}\| = 3$, $\|\vec{v}\| = 4$ and $\|\vec{u} + \vec{v}\| = 5$, the product is $\dfrac{25 - 9 - 16}{2} = 0$. We deduce orthogonality without coordinates.

Watch out

In general, $\|\vec{u} + \vec{v}\| \neq \|\vec{u}\| + \|\vec{v}\|$. The identities concern the squares of the norms.

Immediate application. We are given $\|\vec{u}\| = 2$, $\|\vec{v}\| = 3$, $\vec{u} \cdot \vec{v} = 1$. Calculate $\|\vec{u} - \vec{v}\|$.

3.4 Al-Kashi's theorem and choosing a method

Definition and results

In a triangle ABC, $BC^2 = AB^2 + AC^2 - 2AB\,AC\cos\widehat{BAC}$. This formula is used to calculate the third side from two sides and the included angle, or to determine the cosine of an angle from the three sides.

Status and justification. By Chasles, $\overrightarrow{BC} = \overrightarrow{AC} - \overrightarrow{AB}$. Expanding its scalar square gives $BC^2 = AC^2 + AB^2 - 2AB \cdot AC$; introduce the expression with the cosine.

Worked example. $AB = 5$, $AC = 3$ and $\widehat{BAC} = 60°$: $BC^2 = 25 + 9 - 15 = 19$, so $BC = \sqrt{19}$. With three sides, isolate the cosine before approximating the angle on a calculator, if allowed.

Watch out

The angle is the one between the two sides in the product term. The minus sign is essential.

Immediate application. A triangle has $AB = AC = 4$ and $BC = 4\sqrt{2}$. Determine angle A.

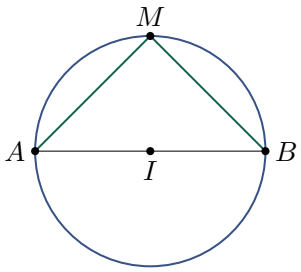
3.5 The circle with diameter AB

Definition and results

For two distinct points A and B with midpoint I, $\overrightarrow{MA} \cdot \overrightarrow{MB} = MI^2 - \dfrac{AB^2}{4}$. The set of points M such that $\overrightarrow{MA} \cdot \overrightarrow{MB} = 0$ is the circle with centre I and radius $\dfrac{AB}{2}$, points A and B included.

Status and justification. Write $\overrightarrow{MA} = \overrightarrow{MI} + \overrightarrow{IA}$ and $\overrightarrow{MB} = \overrightarrow{MI} - \overrightarrow{IA}$. The cross terms cancel ; leaving $MI^2 - IA^2$. Since $IA = \dfrac{AB}{2}$, setting it equal to zero is equivalent to the metric equation of the circle.

Worked example. $A(-2; 0)$ and $B(2; 0)$: I is the origin. The locus is $x^2 + y^2 = 4$. For $M(0; 1)$, the product is $1 - 4 = -3$; for $M(0; 2)$, it equals 0.



Circle with diameter AB: the vectors $\overrightarrow{MA}$ and $\overrightarrow{MB}$ are orthogonal.

Watch out

The endpoints belong to the locus even though "right-angled triangle" does not apply to a degenerate triangle.

Immediate application. For $A(0;0)$, $B(6;0)$, find the points M such that $\overrightarrow{MA}\cdot\overrightarrow{MB}=7$.

Exercise path

Start with exercises 1 and 2. Then choose exercise 3 or 4 without help. Exercises 5 to 7 are a bank of transfer, investigation or programming tasks: the teacher assigns one. Exercise 8 is revisited later.

Exercise 3.1 · Direct calculations

In an orthonormal basis of the plane, consider the vectors $\vec{u}=(1;-2)$ and $\vec{v}=(4;3)$.

1. Calculate the scalar product $\vec{u}\cdot\vec{v}$, then the norms $\|\vec{u}\|$ & $\|\vec{v}\|$.
2. Calculate the coordinates of $\vec{u}+\vec{v}$ and deduce $\|\vec{u}+\vec{v}\|^2$.
3. Recover this last result using the identity involving $\vec{u}\cdot\vec{v}$.

Exercise 3.2 · A right-angled triangle?

The plane has an orthonormal frame. We are given the points $A(1;1)$, $B(4;2)$ and $C(0;4)$.

1. Calculate the coordinates of $\overrightarrow{AB}$ and $\overrightarrow{AC}$, then their scalar product.
2. Deduce that the triangle ABC is right-angled; state at which vertex.
3. Calculate the lengths of the two perpendicular sides, then the area of the triangle in units of area.

Exercise 3.3 · Length without coordinates

The quadrilateral $ABCD$ is a parallelogram such that $AB=4$, $AD=3$ and $\widehat{BAD}=60°$. The lengths are expressed in the same unit.

1. Draw a figure and calculate the scalar product $\overrightarrow{AB}\cdot\overrightarrow{AD}$.
2. Determine the exact lengths of the diagonals $[AC]$ and $[BD]$. Explain how the vectors corresponding to the diagonals are expressed in terms of $\overrightarrow{AB}$ and $\overrightarrow{AD}$.
3. Say which diagonal is longer and check that this result is consistent with your figure.

Exercise 3.4 · Find an angle

Two non-zero vectors $\vec{u}$ and $\vec{v}$ satisfy $\|\vec{u}\|=2$, $\|\vec{v}\|=6$ and $\vec{u}\cdot\vec{v}=-6$. Let θ be their non-oriented angle, between $0°$ and $180°$.

1. Determine the exact value of $\cos\theta$, then the measure of θ.

2. Explain how the sign of the scalar product predicted that the angle would be obtuse.

Exercise 3.5 · A variable locus

The points A and B are distinct and fixed. Let I be midpoint of $[AB]$. For a real k, we look for the set of points M of the plane that satisfy $\overrightarrow{MA} \cdot \overrightarrow{MB} = k$.

1. Express this scalar product in terms of MI and AB using Chasles' relation and a scalar identity.
2. Describe precisely the required set according to the value of k. For each case, indicate whether it is empty, a single point or a circle; then give its characteristics.
3. Explain the role of the fact that a squared distance is positive or zero.

Exercise 3.6 · Work done by a force

An object moves in a straight line from A to B over a distance of 4 m. It is subjected to a force constant $\vec{F}$ of magnitude 50 N. The work done by this force is defined by $W = \vec{F} \cdot AB$ and is expressed in joules.

1. Calculate W when the angle between $\vec{F}$ and $\overrightarrow{AB}$ is 60 °.
2. Repeat the calculation for an angle of 120 °.
3. Compare the signs of the two amounts of work and explain, in each situation, whether the force helps the motion or opposes it.

Exercise 3.7 · Contrapositive

We want to prove the following statement, valid for two vectors in the plane:

$$\|\vec{u}+\vec{v}\|^2 \neq \|\vec{u}\|^2 + \|\vec{v}\|^2 \quad \Longrightarrow \quad \vec{u} \text{ and } \vec{v} \text{ are not orthogonal} \quad .$$

1. State the contrapositive of this implication.
2. Prove this contrapositive using a scalar identity, then conclude about the original statement.

Exercise 3.8 · Delayed review

A triangle ABC satisfies $AB = 3$, $AC = 4$ and $BC = 5$. We seek $\overrightarrow{AB} \cdot \overrightarrow{AC}$ without using any coordinates.

1. Calculate this scalar product by a formula involving the three lengths.
2. Find the result again by determining the nature of the triangle. State the theorem used and its numerical condition.

Independent work. Do only the task assigned at the end of the session, as set in the dated scheme of work: 10-20 min maximum. The exercises depend on the concepts already studied; some questions are therefore postponed. Revisit exercise 8 one week after the chapter (10 min). If the planned time is exceeded, note the difficulty and bring your attempts; do not try to do the whole reserve.

To be done in the CQFD textbook

These exercises are done in your textbook. The scheme of work gives the session; they replace the homework planned for that day (10 to 20 minutes in total).

CQFD, exercise 9 p. 250, questions a and b. Calculate the scalar product in the orthonormal basis and detail both products of coordinates.

CQFD, exercise 11 p. 250, questions 1, 2 and 3. Use in turn the coordinates, the Chasles' relation and the fact that I is the midpoint.

MCQ 1: understand and choose

Five teaching questions with a single answer. Choose one answer per question; then explain one distractor. This chapter format does not reproduce the official format of the EAM.

1. $(2;1)\cdot(3;-4)$ equals

   A. 2

   B. 10

   C. $(6;-4)$

   D. -2

2. The norm of (-3; 4) is

   A. 7

   B. 25

   C. 5

   D. 1

3. Two vectors with zero dot product are

   A. equal

   B. collinear

   C. opposite

   D. orthogonal

4. Norms 2 and 3, angle 60 $^\circ$: product

   A. 6

   B. 3

   C. $\sqrt{3}$

   D. 0

5. $\|u-v\|^2$ is

   A. $\|u\|^2-\|v\|^2$

   B. $\|u\|^2+\|v\|^2$

   C. $\|u\|^2-2u\cdot v+\|v\|^2$

   D. $\|u\|^2+2u\cdot v+\|v\|^2$

MCQ 2: understand and choose

Five teaching questions with a single answer. Choose one answer per question; then explain one distractor. This chapter format does not reproduce the official format of the EAM.

1. An obtuse angle between non-zero vectors gives a dot product

   A. positive

   B. zero

   C. equal to 1

   D. negative

2. In an orthonormal basis, $u\cdot i$ is

   A. the x-coordinate of u

   B. the norm of u

C. the y-coordinate of u

D. always zero

3. The locus $\overrightarrow{MA} \cdot \overrightarrow{MB} = 0$ for $A \neq B$ is

A. perp. bisector

B. the line AB

C. the circle with diameter AB

D. the midpoint alone

4. If $u \cdot v = 3$, $(2u) \cdot (-v)$ is

A. -3

B. -6

C. 6

D. 9

5. For $AB = AC = 2$ and a right angle at A, BC is

A. 4

B. 2

C. 8

D. $2\sqrt{2}$

True or false? Justify.

1. The scalar product of two vectors is always positive.

2. The zero vector is orthogonal to every vector.

3. $u \cdot v = 0$ implies that one of the vectors is zero.

Personal review

I distinguish norm and squared norm; I choose a suitable formula; I check the coordinate system; I write an equivalence for a locus.
For each objective, indicate: mastered without help; achieved with help; needs more work. Write an example and a mistake to avoid.

Solutions and teaching notes, chapter 3

Teacher use. The following answers concern the tasks not solved in the lesson. First have pupils make their approach explicit; provide graduated help before the solution. The durations are estimates to be calibrated in class.

Diagnostic and remediation

1. $(3; -4)$. *Action:* Subtract start from end on two drawings.
2. $\sqrt{3^2 + (-4)^2} = 5$. *Action:* Review Pythagoras and distinguish between norm and coordinates.
3. $\overrightarrow{AC}$. *Action:* Draw the combined displacement.
4. $a^2 + 2ab + b^2$. *Action:* Identify the cross term.
5. $\dfrac{3}{5}$. *Action:* Name the sides with respect to the angle.
6. $(1; 2)$. *Action:* Average the two coordinates separately.

Investigation

Take points on the circle with diameter AB; proof using the midpoint I: product $MI^2 - 9$. The endpoints are included because one of the vectors is zero.

Direct applications

1. 0 then -10. The cosine is respectively 0 and -1.
2. $3t - 12 = 0$, therefore $t = 4$.
3. The square equals $4 - 2 + 9 = 11$; the norm is $\sqrt{11}$.
4. $32 = 16 + 16 - 32\cos A$, therefore $\cos A = 0$ and $A = 90°$.
5. $MI^2 - 9 = 7$: circle with centre (3; 0) and radius 4.

Exercises: methods and answers

Exercise 3.1 - Direct calculations. Product $4 - 6 = -2$; norms $\sqrt{5}$ and 5; sum (5; 1) with square 26. Check: $5 + 2(-2) + 25 = 26$.

Exercise 3.2 - A right-angled triangle? $\overrightarrow{AB} = (3; 1)$, $\overrightarrow{AC} = (-1; 3)$; product 0. $AB = AC = \sqrt{10}$, area $\dfrac{AB \times AC}{2}$ = 5 square units.

Exercise 3.3 - Length without coordinates. $\overrightarrow{AC} = \overrightarrow{AB} + \overrightarrow{AD}$ and $\overrightarrow{BD} = \overrightarrow{AD} - \overrightarrow{AB}$. Product $\dfrac{4 \times 3}{2} = 6$; $AC^2 = 16 + 9 + 12 = 37$ and $BD^2 = 16 + 9 - 12 = 13$.

Exercise 3.4 - Finding an angle. $\cos\theta = -\dfrac{6}{12} = -\dfrac{1}{2}$; in [0; 180 °], the angle is 120°. The domain ensures uniqueness.

Exercise 3.5 - A variable locus. $MI^2 = k + \dfrac{AB^2}{4}$. If the right-hand side is negative: empty set; zero: the single point I; positive: circle whose radius is its square root. The discussion concerns a squared distance.

Exercise 3.6 - Work done by a force. $W = \dfrac{50 \times 4 \times 1}{2} = 100$ J. À $120°$, $W = -100$ J. The sign distinguishes helping the motion from opposing it; it does not describe a negative distance.

Exercise 3.7 - Contrapositive. The direct statement "orthogonal implies equality of squares" follows from the expansion. Its contrapositive is exactly the statement. One can also write 2 $\vec{u} \cdot \vec{v} \neq 0$.

Exercise 3.8 - Delayed review. Al-Kashi: $\dfrac{9 + 16 - 25}{2}$ = 0. Converse of Pythagoras' theorem: triangle right-angled at A, therefore zero product. Name the assumptions of each method.

CQFD, exercise 9 p. 250, questions a and b. a : $1 \times 3 + 2 \times (-1) = 1$. b : $(-1) \times 12 + 3 \times 7 = 9$. The scalar product is a number, not a vector.

CQFD, exercise 11 p. 250, questions 1, 2 and 3. $\overrightarrow{AB} = (4; 6)$, $\overrightarrow{AC} = (-3; 4)$, $I(-1; -2)$ and $\overrightarrow{CI} = (5; -1)$. Hence $\overrightarrow{AB} \cdot \overrightarrow{AC} = 12$, $\overrightarrow{AB} \cdot \overrightarrow{CI}$ = 14. Chasles' relation gives $\overrightarrow{AB} \cdot \overrightarrow{AI}$ = 12 + 14 = 26. Two checks: $\overrightarrow{AI}$ = (2; 3) gives 8 + 18 = 26; or $\overrightarrow{AI} = \dfrac{1}{2}\overrightarrow{AB}$ gives $\dfrac{1}{2}(16 + 36) = 26$.

MCQ: answers and reasons

MCQ 1.

1. A. $6 - 4 = 2$.

2. C. Square root of 25.

3. D. Scalar criterion.

4. B. $\dfrac{2 \times 3}{2} = 3$.

5. C. Expand the scalar square.

MCQ 2.

1. D. Negative cosine.

2. A. Coordinates by projection.

3. C. $MI^2 = \dfrac{AB^2}{4}$.

4. B. Bilinearity: 2($-1)3$.

5. D. $BC^2 = 8$.

True or false

1. False: $u \cdot (-u) = -\|u\|^2 < 0$ for $u \neq 0$.

2. True: its product with any vector is 0.

3. False: (1; 0) and (0; 1) are non-zero and orthogonal.

Differentiated teaching. If a pupil is stuck on a technical point, suggest only the first step (domain, factor or reference sample space), then remove the help. For an independent student, ask for a second method and the handling of a boundary case. Orally, allow an initial rephrasing in the language the pupil is fluent in, then require a precise mathematical sentence in French.